Experiment 25

Title

Write a Prolog Program to implement the Monkey Banana Problem.

Aim

To implement the Monkey Banana Problem using Prolog and determine the sequence of actions required for the monkey to obtain the banana.

Objective

To understand how Prolog uses facts and rules to solve a state-space problem by finding the required sequence of actions.

Algorithm

Step 1: Define the initial state of the monkey, box, and banana.

Step 2: Define the final state where the monkey has obtained the banana.

Step 3: Define the possible actions and their effects on the current state.

Step 4: Execute the actions recursively until the goal state is reached.

Step 5: Display the sequence of actions required to solve the problem.

Program

```
% Initial and Final States


initial_state(state(at_door, on_floor, at_window, has_not_eaten)).

final_state(state(_, _, _, has_eaten)).


% Actions


action(state(at_door, on_floor, at_window, has_not_eaten),

    climb,

    state(at_window, on_window, at_window, has_not_eaten)).


action(state(at_window, on_window, at_window, has_not_eaten),

    grasp,
```

```
    state(at_window, on_window, at_window, has_eaten)).

% Execute Actions

execute([], State, State).

execute([Action|Rest], CurrentState, FinalState) :-
  action(CurrentState, Action, NextState),
  execute(Rest, NextState, FinalState).

% Solve Problem

solve(Actions) :-
  initial_state(State),
  execute(Actions, State, FinalState),
  final_state(FinalState).
```

Sample Input

?- solve(Actions)

Sample Output

```
consult('C:/Users/B SANTHOSH KUMAR/OneDrive/Documents/exp25.pl').
compiling C:/Users/B SANTHOSH KUMAR/OneDrive/Documents/exp25.pl for byte code...
C:/Users/B SANTHOSH KUMAR/OneDrive/Documents/exp25.pl compiled, 28 lines read - 2564 bytes written, 4 ms

yes
| ?- solve(Actions).

Actions = [climb,grasp] ?

yes
```

Result

Hence, the Prolog program to implement the **Monkey Banana Problem** was executed successfully and the correct sequence of actions was obtained.